

Personal Reflection

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1. What I set out to do

I started this project with a problem that is easy to state and hard to attack: the human genome admits on the order of 10^8 candidate gene pairs, synthetic lethality (SL) is context-specific, and combinatorial CRISPR screens cannot cover that space. The clinical success of PARP inhibitors in BRCA-mutant tumors told me the payoff is real; the scale told me that computational prioritization is a prerequisite, not a luxury. So I narrowed the question to something I could actually defend: within one well-characterized context (K562), can I rank a gene's candidate SL partners?

The starting observation that carried the whole project was this: when a gene is silenced, the cell's genome-wide transcriptional response — the perturbation “shockwave” — reflects that gene's functional wiring. If that same response can predict a gene's *own* dependency, it should also carry information about the partners whose loss it cannot tolerate. The entire arc of my work was an attempt to test that single idea honestly, and then to push it past the genes I could actually measure.

2. The methodology I built, and why I built it that way

Looking back, the thing I am most satisfied with is not any single model — it is the *ladder of evidence* I forced myself to climb in order, instead of jumping straight to the deep-learning centerpiece.

- **Foundation first (exp01–exp02).** Before claiming anything about partners, I proved the prerequisite: pseudobulk Δ -expression regressed onto DepMap GeneEffect reached ~ 0.49 Spearman in cross-validation and transferred to Adamson at AUROC 0.886. Crucially, I then audited it — was the model just reading a generic “everything dies” viability axis? The death signature alone reached only 0.244, and residualizing it out left the transcriptomic signal essentially intact at 0.503. That audit mattered more to me than the headline number, because it is the difference between a real result and a comfortable illusion.
- **A representation as the bridge (exp03–exp05).** I moved from pseudobulk to single-cell set learning, where a frozen-GMM distribution regressor on scVI embeddings won (Adamson Spearman 0.666), and then connected the frozen STATE forward model. This gave me a proposed route for turning a perturbation into a downstream dependency feature — the capability I needed to attempt the cold-start frontier, even though exp08 later showed that this route was not yet strong enough.

- **A floor before a ceiling (exp06).** I built the most boring possible baseline first: just the two genes' DepMap GeneEffect scalars, five swap-invariant features, predict P(SL). It turned out to be strong (CV2 AUROC 0.704). That floor became the bar every later, fancier model had to clear.
- **The proof-of-concept (exp07).** Holding the harness fixed and changing only the feature matrix, adding the observed transcriptome more than doubled CV2 ranking quality (NDCG@10 0.042 $\rightarrow$ 0.094). This was the empirical license for everything that followed.
- **The generative framework (exp08) and a parallel route (exp09).** Finally the e2e model that *generates* the response for unscreened genes, plus a cross-cell-line selectivity route that needs no transcriptome at all.

Building in this order taught me that the sequence of experiments is itself an argument. Each rung answered a question that made the next rung interpretable.

3. What I am proudest of: the discipline of honesty

If I had to name the single most valuable habit I practiced, it is refusing to let the framing outrun the evidence. Several decisions encode this:

- **I treated CV1 as a diagnostic, not a result.** A degree-only probe wins CV1 ranking (NDCG@10 0.197) yet collapses to ~ 0.001 on CV2/CV3. So CV1 rewards graph topology, not biology, and I judged every model only on CV2 and CV3 — the genuine generalization surfaces.
- **I scoped the label precisely.** The benchmark D is a SynLethDB-derived adapter target whose negatives are *unconfirmed* non-SL pairs, not a validated K562 SL assay. I never let a high-ranked pair be described as a confirmed target. It is candidate prioritization, full stop.
- **I kept the comparison fair.** The published methods (DDGCN, GRSMF, SL2MF, SLGNN) lead on ranking, and I said so plainly rather than hiding behind within-harness lifts. The honest causal claims came only from same-harness ablations (exp06 vs exp07 vs exp09).

4. The hardest technical problems

The closed-vocabulary bottleneck was the defining engineering challenge. My local STATE checkpoint identifies a perturbation through a one-hot lookup over a fixed gene set, and only 16.3% of my 9,471-gene universe was in that vocabulary. For the cold-start splits, 84% of genes would simply get no signal. My fix — freeze the STATE backbone and replace its one-hot perturbation encoder with a trainable adapter fed by ESM2 protein embeddings — was the moment the project went from “interesting baseline study” to “a method.” Mapping every gene into one continuous coordinate system, regardless of whether it was ever screened, is what gives the approach any path to held-out genes at all.

The other recurring difficulty was *leakage control*. Making CV2/CV3 valid meant held-out genes had to be reached purely through adapter (ESM2) + frozen STATE, and bag supervision had to be restricted to training genes that had a real profile. It would have been easy to accidentally let a held-out gene see its own observed response and quietly inflate the numbers. Designing the three-part loss and the data flow so that this *could not* happen took more care than the model architecture

itself.

5. What did not work, and what I learned from it

The part that most clearly did not work was the exp08 end-to-end generative pipeline. I built it for a real reason: observed Perturb-seq profiles helped in CV2, but they cannot cover the whole SL universe, so the next logical step was to generate the missing shockwave with frozen STATE plus an ESM2-conditioned adapter. The current artifacts show that this idea is not yet successful. The result set itself is incomplete — there is no top-level official summary, and several runs have epoch CSVs without final fold-result JSONs — so I cannot call it a finished official result. But the completed folds are already negative: Phase 2 CV2 reaches only AUROC/AUPR 0.6667/0.6754 and NDCG@10/MAP@10 0.0050/0.0035, far below the exp06 XGBoost floor of 0.7035/0.7323 and 0.0421/0.0341. Phase 3, where I added real GWPS bag supervision, does not fix this; selected CV2 gets worse, and the raw five-fold CV2 mean is still near random-ranking behavior (AUROC 0.5315, NDCG@10 0.0069).

That failure taught me not to confuse “uses richer biological data” with “preserves useful biological signal.” The logs suggest the model was not simply crashing: Phase 3 runs applied optimizer steps without skipped updates. The problem is that the learned signal was weak. Validation AUROC often peaked early and then fell back toward 0.5, while the Phase 3 loss scale was dominated by the bag objective around 11–12 without producing better SL ranking. My interpretation is that the frozen STATE + small adapter path was too constrained for this benchmark: the adapter, pooling, and pair head were trainable, but the backbone was frozen; bag supervision was available only for covered training genes; and the main optimization was still mostly pair-level BCE even though the paper’s real metric is per-anchor top-k ranking. In other words, the pipeline could run, but it did not learn a stable ordering of synthetic-lethal partners.

This negative result does *not* mean real perturbation data is useless. Exp07 showed the opposite: direct real-bag features substantially improved CV2 ranking when one partner was anchored by a measured profile. What failed in exp08 was the translation step from measured response signal into a generated representation through a frozen forward model. That distinction matters because it tells me what to debug next: ranking-aligned objectives, loss balancing, response-fidelity diagnostics, and possibly a less frozen forward model, rather than simply adding more GWPS supervision and hoping the signal appears.

Cold-start ranking (both genes unseen, CV3) also remained unsolved by every signal I tested. The decomposition experiment explained why, and this was probably my most important negative finding: when I restricted CV3 to non-pan-essential pairs, performance fell back toward chance. Most of the genome-wide co-dependency signal is *pan-essentiality* — broadly essential genes are easy to flag — and pair-specific synthetic lethality is the thin, biologically interesting residual that hides underneath. That reframed the whole problem for me. The cold-start gap is not just a tuning issue; it is a precise scientific target. A generative model has to learn the pair-specific signal that essentiality gravity obscures, and my current exp08 pipeline has not done that yet.

Learning to value a clean negative result as much as a positive one was a genuine shift in how I think about research.

6. What this taught me

- **Evaluation design is the real research.** Choosing CV2/CV3 over CV1, building a strong floor, and auditing for confounds shaped my conclusions more than any single model did.
- **Simple baselines are not a formality.** The dependency-only floor was strong enough to expose when a deep model was not actually learning useful ranking signal, which is exactly why it was worth building first.
- **Foundation models are powerful but constrained.** A frozen backbone plus a small trainable adapter is data-efficient and respects a pretrained prior, but it also creates a hard bottleneck: if the adapter cannot drive the frozen model into the right perturbation regime, richer supervision will not automatically become better downstream ranking.

7. Where I would go next

The natural next steps follow directly from the limitations I documented:

1. Complete a clean five-fold exp08 evaluation with a validation split that does *not* select on the test fold, so the generative framework can be judged as a real comparison rather than a preliminary artifact set.
2. Align training with the actual objective by adding per-anchor ranking losses, rebalancing the SL and bag-supervision terms, and diagnosing generated-response fidelity before using those responses for SL prediction.
3. Test whether a less constrained adapter or partially trainable forward model can preserve the real-bag signal that exp07 showed exists.
4. Push beyond K562 — cross-lineage transfer, and eventually patient-context mapping from cell line dependency to tumor prioritization.
5. Supervise the model directly on the pair-specific residual that pan-essentiality otherwise hides, since that is where the genuinely useful SL signal lives.

8. Closing

I proved that the observed transcriptome encodes a gene's partners when one gene is anchored by a real profile, I built a generative framework that can in principle reach the 83.7% of genes outside the foundation model's vocabulary, and I learned that this framework does not yet preserve the signal needed for synthetic-lethal top-k ranking. For a problem this large and this prone to self-deception, knowing the boundary of my own claims feels like the most durable thing I produced.